AlphaProteo Reading Summary

Lou Hongfu

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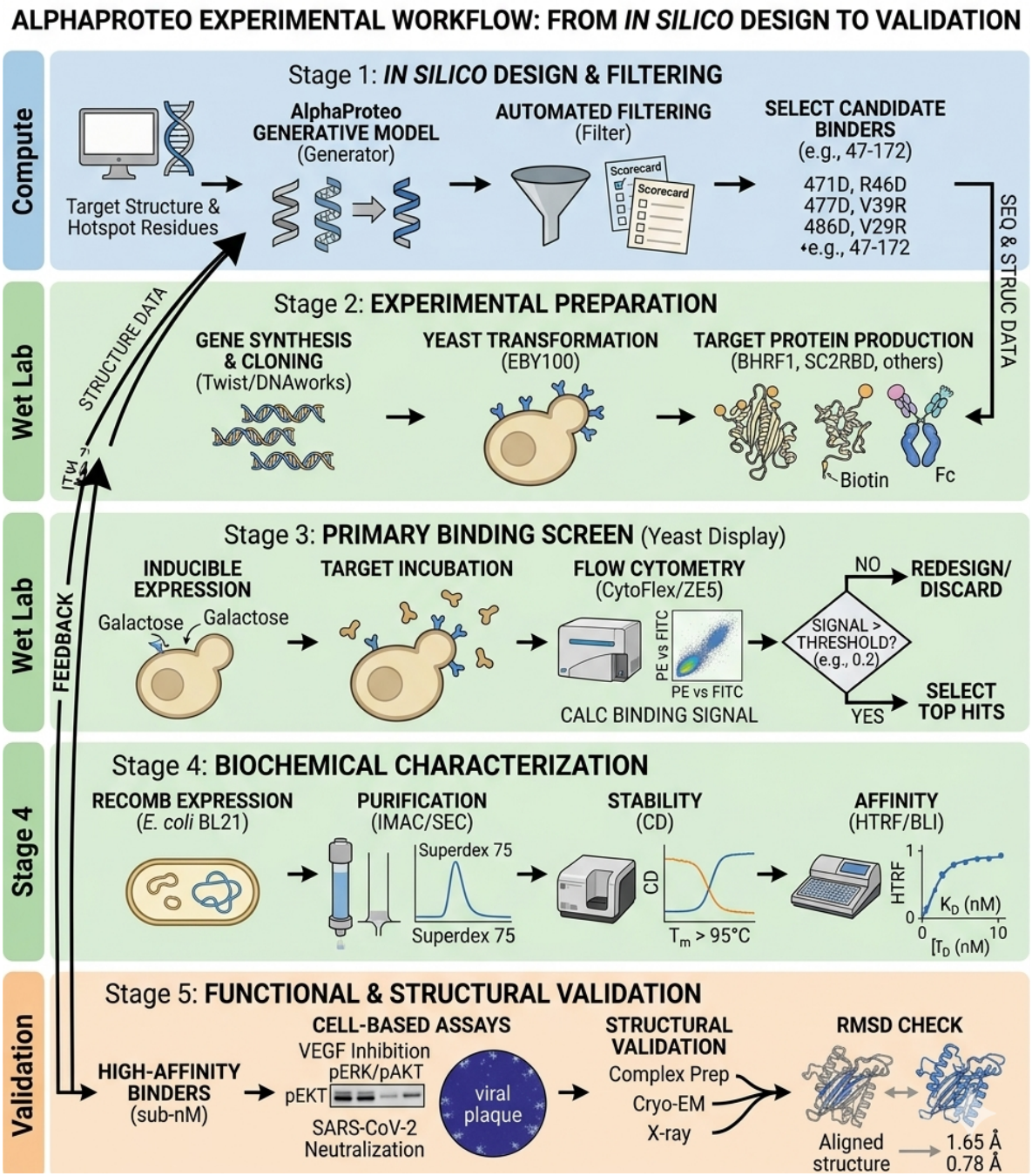

Figure 1: Workflow(generated by Gemini)

1 Introduction

Because AlphaProteo is not open source until now, I may talk more about the experimentation part of this paper (or report) on Arxiv `/https://arxiv.org/abs/2409.08022`.

In these paper, they choose 8 kinds of specific protein to design their binder using AlphaProteo. After designing, they need to validate the whether to combine, the binding affinity, the conformation and the thermal stability.

For binder purification, I plan to skip because there aren't too many physical principles involved.

BHRF1, SC2RBD, IL-7RA, PD-L1, TrkA, IL-17A, VEGF-A, TNFα

2 Target protein expression and purification

For IL-7RA(21-239), TrkA(34-423), PD-L1(19-239), VEGF-A(27-191), IL-17A(24-155) and VEGF165, they purchased them from other company.

2.1 Non-Equilibrium Nature

Protein expression is not an equilibrium process: it is continuously driven by energy (NTP/ATP) flux. A simple kinetic description is:

$$\frac{d[P]}{dt} = k_{\text{prod}} - k_{\text{deg}}[P]$$

Here, k_{prod} depends on transcription and translation rates which can be controlled by IPTG, while k_{deg} reflects cellular degradation.

Intuition: The cell maintains a steady-state protein concentration, not a thermodynamic equilibrium. Energy input keeps the system dynamically active.

2.2 IPTG Modulates Transcription Probability

IPTG works by binding LacI, changing its conformation, and reducing its affinity for DNA:

$$P_{\text{DNA unbound}} = \frac{1}{1 + e^{-\Delta G/kT}}$$

The transcription rate scales with the probability that the promoter is free:

$$k_{\text{transcription}} \propto P_{\text{DNA unbound}}$$

Intuition: IPTG does not just "switch on" transcription; it tunes the promoter occupancy continuously, effectively setting k_{prod}.

2.3 Temperature and Folding Kinetics

After translation, proteins can fold correctly or aggregate:

$$\text{nascent protein} \begin{cases} \xrightarrow{k_{\text{fold}}} \text{native} \\ \xrightarrow{k_{\text{agg}}} \text{aggregate} \end{cases}$$

Temperature affects both translation and folding:

- High T: faster translation $\rightarrow$ less time for proper folding $\rightarrow$ misfolding increases
- High T: conformational fluctuations $\uparrow \rightarrow$ aggregation increases
- Low T: slower translation $\rightarrow$ more time for co-translational folding $\rightarrow$ aggregation decreases

Functional protein yield:

$$[P_{\text{functional}}] \propto k_{\text{prod}} \cdot \frac{k_{\text{fold}}}{k_{\text{fold}} + k_{\text{agg}}}$$

Intuition: Temperature shifts the balance of flux through folding vs aggregation pathways, not just "slows things down".

2.4 Unified Kinetic Picture

A complete view combines production, folding, and aggregation:

$$\begin{cases} \frac{d[P_{\text{nascent}}]}{dt} = k_{\text{prod}}(\text{IPTG}, T) - (k_{\text{fold}} + k_{\text{agg}})[P_{\text{nascent}}] \\ \frac{d[P_{\text{functional}}]}{dt} = k_{\text{fold}}[P_{\text{nascent}}] - k_{\text{deg}}[P_{\text{functional}}] \\ \frac{d[P_{\text{aggregate}}]}{dt} = k_{\text{agg}}[P_{\text{nascent}}] \end{cases}$$

Intuition: The final functional protein concentration emerges from a competition between production, folding, and irreversible aggregation, modulated by IPTG and temperature.

3 Binding Detection as a Signal Transduction System

3.1 From Molecular Binding to Observable Signal

The goal of the assay is to convert an invisible molecular binding event into a measurable fluorescence signal. This is achieved through a cascade:

$$\text{binder} + \text{target} \rightleftharpoons \text{complex}$$

$$\text{complex} \xrightarrow{\text{biotin-streptavidin}} \text{fluorescent signal (PE)}$$

Intuition: Binding itself is not directly observable. The system introduces a molecular "linker" (biotin) and a fluorescent reporter (PE) to transform binding into light.

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3.2 Fluorescence as a Proxy for Physical Quantities

Two fluorescence channels are measured:

$$F_{\text{FITC}} \propto [\text{binder expression}]$$

$$F_{\text{PE}} \propto [\text{binder-target complex}] + F_{\text{background}}$$

Intuition: FITC measures how much binder is present, while PE measures how much target is captured. The experiment separates expression from binding.

3.3 Binding Equilibrium and Signal Generation

Binding between binder (B) and target (T) can be described as:

$$B + T[k_{\text{off}}]k_{\text{on}}BT$$

At quasi-steady state:

$$[BT] \propto \frac{[B][T]}{K_D}$$

The measured fluorescence becomes:

$$F_{\text{PE}} \sim [BT] + F_{\text{non-specific}}$$

Intuition: Strong binders (low K_D) produce more complexes and therefore stronger fluorescence. However, background signals are always present and must be corrected.

3.4 Background Subtraction and Signal Isolation

To isolate true binding, four populations are considered:

$$\text{signal} = (\log_{10} F_{\text{FITC+,+target}} - \log_{10} F_{\text{FITC-,+target}}) - (\log_{10} F_{\text{FITC+,-target}} - \log_{10} F_{\text{FITC-,-target}})$$

Intuition: The first term captures total binding-related signal, while the second term removes contributions from non-specific interactions and expression-dependent artifacts. The result is a net binding signal.

3.5 Signal Decomposition

The observed fluorescence can be decomposed as:

$$F_{\text{PE}} = F_{\text{specific}} + F_{\text{non-specific}} + F_{\text{expression-dependent}}$$

Thus:

$$\text{binding signal} \approx F_{\text{specific}}$$

Intuition: The analysis pipeline removes all contributions except the fluorescence arising directly from binder-target interaction.

3.6 Unified Detection Model

Combining expression, binding, and detection:

$$\begin{cases} [B] \sim \text{expression level (FITC)} \\ [BT] \sim \frac{[B][T]}{K_D} \\ F_{\text{PE}} \sim [BT] + \text{background} \\ \text{signal} \sim \log([BT]) - \text{background correction} \end{cases}$$

Intuition: The final measured signal is not a direct readout of binding, but the result of a multi-step transformation from molecular interaction to fluorescence, followed by statistical correction.

4 Interface Validation, Competition, and Specificity Assays

4.1 Interface Mutation as a Perturbation Test

To validate whether the predicted binding interface is correct, mutations are introduced at the binder–target interface:

$$\text{binder}_{\text{WT}} \rightarrow \text{binder}_{\text{mut}}$$

Specifically, hydrophobic residues at the interface are replaced with charged residues:

$$\text{hydrophobic (A, V, L, I)} \;\rightarrow\; \text{charged (D, E, R, K)}$$

Binding is then re-measured using the same fluorescence-based assay:

$$F_{\text{PE}}^{\text{mut}} \propto [\text{binder}_{\text{mut}} \cdot \text{target}]$$

Experiment: Mutant binders are expressed on yeast and subjected to the same binding screen as the wild-type designs.

Intuition: Hydrophobic residues often stabilize protein–protein interfaces. Replacing them with charged residues disrupts packing and introduces electrostatic penalties. If binding decreases significantly, it indicates that the mutated residues are part of the true interface.

4.2 Competition Assay for Binding Site Overlap

To determine whether the binder targets a specific site, a competitor protein with known binding to the same target is introduced:

$$\text{binder} + \text{target} + \text{competitor}$$

Binding becomes a competitive process:

$$\text{target} + \text{binder} \rightleftharpoons BT$$

$$\text{target} + \text{competitor} \rightleftharpoons CT$$

The observed binding signal:

$$F_{\text{PE}} \propto [BT]$$

Experiment: Yeast-displayed binders are incubated with biotinylated target in the presence or absence of a competitor protein. Binding is measured via streptavidin-PE fluorescence as before.

Intuition: If the binder and competitor share the same binding site, the competitor reduces the available free target, decreasing $[BT]$ and thus the PE signal. If the signal remains unchanged, the two likely bind distinct sites.

4.3 Avidity-Enhanced Detection for Weak Binding

For weak interactions, multivalent complexes are formed prior to binding:

$$\text{target} + \text{detection reagents} \rightarrow \text{multivalent complex}$$

Experiment: Target proteins are pre-incubated with fluorescent detection components before being added to cells.

Intuition: Multivalent interactions increase effective binding strength (avidity), stabilizing otherwise transient interactions and enhancing detectable signal.

4.4 Specificity Measurement via HTRF

Binding specificity is assessed using a homogeneous time-resolved fluorescence (HTRF) assay:

$$F_{\text{HTRF}} \propto [\text{binder} \cdot \text{target}]$$

The assay is performed at fixed concentrations:

$$[\text{binder}] = 1\,\text{nM}, \quad [\text{target}] = 100\,\text{nM}$$

Experiment: Each binder is tested against its intended target (and potentially other proteins) using HTRF, which measures binding via energy transfer between fluorophores.

Intuition: By using low binder concentration and excess target, the assay emphasizes selective binding. A highly specific binder will produce strong signal only with its intended target, while non-specific binders will show broader interactions.

4.5 Integrated Interpretation

The three assays probe complementary aspects of binding:

$$\begin{cases} \text{Mutation} \rightarrow \text{interface validity} \\ \text{Competition} \rightarrow \text{binding site overlap} \\ \text{Specificity} \rightarrow \text{target selectivity} \end{cases}$$

Intuition: Together, these experiments ensure that binding is structurally correct (interface), spatially meaningful (site), and biologically relevant (specificity), rather than arising from non-specific interactions.